

FUELING PERFORMANCE: CALORIES, CARBS, AND HYDRATION FOR ATHLETES

Why carbs, training-day flexibility, and hydration matter more here than in most other goals.

Ask someone training for a sport or an athletic performance goal how many calories they need, and the honest answer is "it depends on the day." That is not a dodge, it is the actual nature of performance-focused eating: the target moves with your training load in a way it does not for most other goals, because the whole point is fueling output, not just managing body weight.

The Baseline: BMR

Basal metabolic rate (BMR) is your energy floor, what your body burns at rest before any training is added. The Mifflin-St Jeor equation estimates it from weight, height, and age:

- **Men:** $BMR = (10 \times \text{weight in kg}) + (6.25 \times \text{height in cm}) - (5 \times \text{age}) + 5$
- **Women:** $BMR = (10 \times \text{weight in kg}) + (6.25 \times \text{height in cm}) - (5 \times \text{age}) - 161$

Take a 22-year-old athlete training for performance, 82 kg (about 181 lbs), 183 cm tall (about 6'0"). His BMR: $(10 \times 82) + (6.25 \times 183) - (5 \times 22) + 5 = 820 + 1,143.75 - 110 + 5 = \mathbf{1,859 \text{ calories}}$.

BMR to TDEE: The Demands of Power and Speed Training

Multiply BMR by an activity multiplier for total daily energy expenditure (TDEE): sedentary $\times 1.2$, lightly active $\times 1.375$, moderately active $\times 1.55$, very active $\times 1.725$. A performance program with power, speed, strength, and conditioning work spread across four hard sessions a week, plus sport-specific training, lands at very active. $1,859 \times 1.725 = \mathbf{3,208 \text{ calories}}$, his maintenance number.

Why This Goal Stays Close to Maintenance, With Flex Built In

Unlike weight loss or bulking, performance nutrition generally is not built around a sustained deficit or surplus, the goal is fueling training and recovery well, not changing body weight in a specific direction. Calories typically sit close to that TDEE number, with real flexibility around the training calendar: **roughly 10% more on the hardest training or competition days**, when carbohydrate needs spike, and a touch less on lighter or rest days. For our example, that is a range generally around **2,900 to 3,500 calories** depending on the day, not a single fixed number applied every day of the week regardless of what is on the schedule.

Carbs Do More Work Here Than in Most Other Goals

Carbohydrate is the fuel that powers high-output training, sprinting, jumping, heavy lifting, repeated hard efforts, more directly than any other macronutrient. For a performance athlete, carb intake should scale up on higher-output days and pull back somewhat on lighter ones, generally the largest portion on the plate on a hard training day, closer to a moderate portion on a rest day. Protein stays a consistent palm-sized portion at each meal regardless of the day, since recovery and injury resilience do not take a day off just because training intensity does. Fat fills the remainder, kept moderate rather than either extreme.

Hydration and Electrolytes Matter More Too

Athletes training hard, especially with repeated sprint, agility, or conditioning work, lose meaningful fluid and electrolytes through sweat in a way that a more moderate training program does not demand as much attention to. Consistent water intake through the day, plus attention to sodium and other electrolytes around long or intense sessions, supports the same performance and recovery goals as the calorie and carbohydrate targets above, it is not a separate conversation from fueling, it is part of it.

Where Supplements Fit

Recovery from repeated hard training sessions depends on getting enough protein in consistently, a whey option like **XS™ Whey Protein** can help close that gap around training without adding a lot of food volume when appetite is competing with a busy training schedule. **XS™ Creatine+** is particularly relevant for this goal specifically, as a training-performance and strength-support option tied directly to the power and repeated-effort output this program is built around. Both are available through [the Echelon Goods shop](#).

What This Means for You

- Performance nutrition starts with the same BMR-to-TDEE math as any other goal, then stays close to maintenance instead of a sustained deficit or surplus.
- Build in roughly 10% more calories, mostly from carbs, on your hardest training days, and ease back slightly on lighter days.
- Carbohydrates carry more of the load here than in most other goals, since they directly fuel high-output training.
- Hydration and electrolytes are part of fueling performance, not an afterthought to it.

This article is educational and general in nature, not individualized medical or nutrition advice. If you have a medical condition or are taking medication, talk to your doctor or a registered dietitian before making significant changes to your diet or training load.